Module 5 - Slide Deck Video Transcription

1 Hi, I'm Jordan Reyes from Tallgrass Advisory, a fictional character presented by an AI avatar, on purpose, because this course is about using AI well and being open about it. Welcome to your second rotation: the South Carolina plant, working for Marcus Delacroix in Cost and Planning. This week Marcus has a real decision to make, and your job is to get AI to do the analysis, check it, and then make the call.

2 Here's Marcus's problem. A rental-car fleet company has offered to buy 120 Velox cars at 240 thousand dollars each. That's about 14 percent below the usual price of roughly 279 thousand. The cars would be delivered by December 15, and the customer wouldn't pay for 90 days, instead of the usual 30. The plant is nearly full in the fourth quarter, so some cars would need overtime. Sales wants to say yes. Marcus wants numbers he can trust before he says yes or no.

3 First, the accounting idea, and it's short. The question is whether Velox is better off with the order than without it. To answer it, only count costs that change because of the decision. Accountants call these relevant costs. Materials, assembly labor, and delivery freight for each extra car all change, so they count. But fixed overhead, like supervisors, equipment depreciation, and utilities, is the same whether or not the plant takes the order. Reports often spread it over each car, but the plant pays the same total either way, so it isn't relevant.

4 Next, what each car adds. Take the price and subtract the costs that change per car. What's left is the contribution margin, what each extra car adds toward fixed costs and profit. Two things change the total. Capacity: cars built in unused regular time carry no extra cost, but cars beyond that need overtime, which costs more per car. And side effects: a customer who pays later ties up cash, and a low price offered to one customer can leak and make regular customers ask for lower prices. You won't do this arithmetic by hand. AI will. Your job is to get it to do it right, and to check it.

5 Let me show you with a made-up example, so I don't give away your assignment. Prairie Tool and Die is a small Velox supplier. A customer offers to buy 1,000 steel brackets at 38 dollars each. The regular price is 52. Each bracket costs 31 dollars to build, plus 9 dollars of fixed overhead that the plant pays regardless. Regular time has room for 700 more brackets, so 300 would need overtime at 5 dollars extra each. A manager worries regular customers might ask for a 2 percent price cut if they hear about the discount. I gave Claude

these facts and asked it to explain three options, accept as offered, decline, or counter at 41 dollars, plus a best, base, and worst case.

6 Here's what Claude returned. This is real output. For accepting as offered, it worked out 700 brackets in regular time at 38 minus 31, which is 7 dollars each, 4,900 dollars. Then 300 overtime brackets at 38 minus 36, 2 dollars each, 600 dollars. Total: plus 5,500. Declining adds nothing. Countering at 41 dollars gives 7,000 plus 1,500, which is plus 8,500, but only if the customer agrees, and Claude noted the files don't say whether they would. Notice that it left the 9 dollars of fixed overhead out, and explained that counting it would make the order look like a 2 dollar loss per bracket, which is misleading because that overhead doesn't change.

7 Now the scenarios. This is the second technique: scenario analysis, also called sensitivity analysis or what-if prompting. Why it helps: a single answer hides the risk, and a decision maker needs to know how wrong it could go. Claude's worst case was a 2 percent price cut on all 4,300 regular brackets, 4,472 dollars lost, leaving the order at plus 1,028. It also found the break-even: a price cut of about 2.5 percent would erase the profit. And here is something I like. Claude said the files give no basis for a better outcome, so best and base are the same. It didn't invent a rosy best case. It also listed what isn't in the files, like how likely a price cut is.

8 Now I do what you'll do: I check it myself. Seven dollars times 700 is 4,900. Two dollars times 300 is 600. Together 5,500. Claude's number matches. I also confirmed the 31 dollar cost against the cost sheet, and that 38,700 of fixed overhead divided by 4,300 brackets is the 9 dollars. Here's why I check the inputs and not just the arithmetic: a correct calculation on the wrong number is still wrong. In your assignment you'll verify two findings this way, using a spreadsheet, and you won't ask AI to check them.

9 First technique: write the request once, as a template with defined inputs. The next special order will have different numbers but the same questions. A template is a request with named blanks in square brackets, and each blank says what it is, its unit, and where to find it. Practitioners call this a parameterized prompt, or a prompt template with variables. Why it helps: next time you only change the inputs, so the request doesn't get sloppy, and defining each input tells you exactly which numbers to look up and check. In your assignment you'll write your own special-order template, with at least six inputs.

10 AI can analyze options and explain their results, but it can't decide, because it doesn't own the consequences. A person does. When you write your recommendation for Marcus, put the recommendation in the first sentence: accept, decline, or counter on stated terms. Then give the few numbers that drive it, what would change your mind, and what you don't

know. In this case, any of the three decisions can earn full credit, as long as your numbers and reasoning support it.

11 This week's ethics checkpoint is area three of the Velox Responsible AI Policy: bias. If the order is accepted, about 70 cars would be built on overtime, and someone has to decide who works it. That's exactly the kind of job where AI can quietly treat people unfairly. A plan that decides who gets an opportunity must not depend on age, health, or family leave, or on a proxy for them. A proxy is a measure that quietly stands in for something you must not use, like attendance records standing in for health. And a person must check the results for patterns. In Step 4 you'll judge five AI suggestions, and you'll describe a bias check of your own.

12 Here's your week. First you'll try the offer in a plain chat. Then you'll learn the ideas and judge five AI suggestions for bias. Then you'll write your own special-order template, run it on the Velox files, check two findings yourself, and make the call in a one-page recommendation for Marcus. You'll also choose a lens, one of six equal options: accounting, finance, marketing, management, supply chain, or information systems. Write your own prompts, check the numbers, and let's get to work.